Tutorial-4 Solutions

Q.1] $P = 8.5\ N$, $E = 3.3\ GPa$, $\varepsilon = \frac{\Delta L}{L} = 1.1\%$

$$\sigma = E\varepsilon = 3.3 \times 10^9 \times 0.011$$

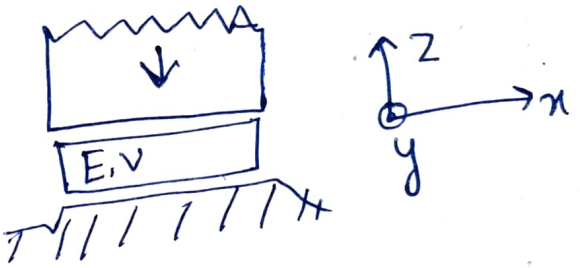
$$\boxed{\sigma = 36.3\ MPa}$$

$$A = \frac{P}{\sigma} = \frac{8.5}{36.3 \times 10^6} = 234.16 \times 10^{-9}\ m^2$$

$$d = \sqrt{\frac{4A}{\pi}}$$

$$\boxed{d = 0.546\ mm}$$

Q.2] $\sigma_z = -P_0$



$$E\varepsilon_x = \sigma_x - \nu\sigma_y + \nu P_0$$

$$E\varepsilon_y = \sigma_y + \nu P_0 - \nu\sigma_x$$

$$E\varepsilon_z = = -P_0 - \nu\sigma_x - \nu\sigma_y$$

a) $\varepsilon_x = \varepsilon_y = 0$

$\therefore 0 = \sigma_x - \nu\sigma_y + \nu P_0$, $0 = \sigma_y + \nu P_0 - \nu\sigma_x$,

$$E\varepsilon_z = -P_0 - \nu\sigma_x - \nu\sigma_y$$

$$\therefore \varepsilon_z = \frac{-1-\nu-2\nu^2}{1-\nu}\frac{P_0}{E}, \quad \sigma_x = \sigma_y = \frac{-\nu}{1-\nu}P_0$$

b) $\varepsilon_y = 0$ & $\sigma_x = 0$

$E\varepsilon_x = -\nu\sigma_y + \nu P_0$, $0 = \sigma_y + \nu P_0$, $E\varepsilon_z = -P_0 - \nu\sigma_y$

$\varepsilon_x = \nu(1+\nu)\frac{P_0}{E}$, $\varepsilon_z = -(1-\nu^2)\frac{P_0}{E}$, $\sigma_y = -\nu P_0$

c) $\sigma_x = \sigma_y = 0$

$E\varepsilon_x = \nu P_0$, $E\varepsilon_y = \nu P_0$, $E\varepsilon_z = -P_0$

$\varepsilon_x = \varepsilon_y = \frac{\nu P_0}{E}$, $\varepsilon_z = -\frac{P_0}{E}$

Q.3] As the block is confined in x direction

$\varepsilon_x = 0$

It is free to move in y direction

$\therefore \sigma_y = 0$

using Hooke's law.

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$0 = \frac{1}{E}(\sigma_x - \nu\sigma_z)$$

$$\sigma_x = \nu\sigma_z$$

similarly

$$\varepsilon_z = \frac{1}{E}(\sigma_z - \nu(\sigma_x + 0)) = \frac{1}{E}(\sigma_z - \nu^2\sigma_z)$$

$$\varepsilon_z = \frac{\sigma_z}{E}(1-\nu^2)$$

The effective modulus of elasticity of the rubber block under the confined condition can be determined by considering the rubber block as unconfined but rather undergoing the same normal strain of ε_z when it is subjected to the normal stress σ_z

$$\therefore \sigma_z = E_{eff}\, \varepsilon_z$$

$$E_{eff} = \frac{\sigma_z}{\varepsilon_z} = \frac{\sigma_z}{\frac{\sigma_z}{E}(1-\nu^2)}$$

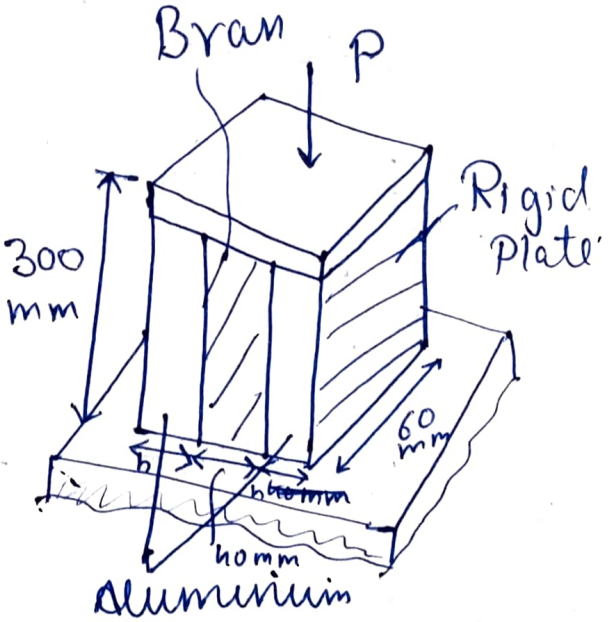
$$\boxed{E_{eff} = \frac{E}{1-\nu^2}}$$

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Let,

P_b = axial force in Bran

P_a = axial force carried by aluminium.



$$\delta = \frac{P_b L}{E_b A_b} \qquad P_b = \frac{E_b A_b \delta}{L}$$

$$P_a = \frac{E_a A_a \delta}{L}$$

$$P = P_a + P_b = (E_b A_b + E_a A_a)\frac{\delta}{L}$$

$$\varepsilon = \frac{\delta}{L} = \frac{P}{E_b A_b + E_a A_a}$$

$A_b = (60)(40) = 2400 = 2400 \times 10^{-6}\ m^2$

$A_a = 2(60)(10) = 1200 = 1200 \times 10^{-6}\ m^2$

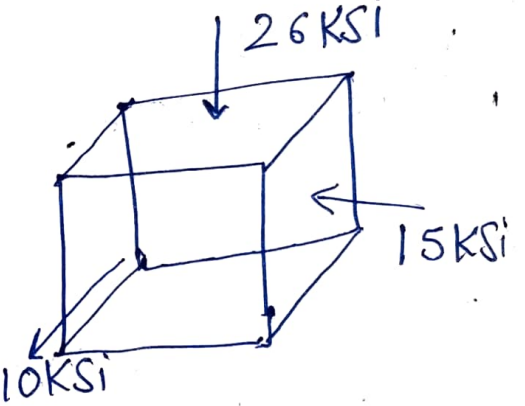
$$\varepsilon = \frac{450 \times 10^3}{(105 \times 10^9)(2400 \times 10^{-6}) + (70 \times 10^9)(1200 \times 10^{-6})}$$

$$\varepsilon = 1.3393 \times 10^{-3}$$

a) $\sigma_b = E_b \varepsilon = (105 \times 10^9)(1.3393 \times 10^{-3}) = 140.6\ MPa$

b) $\sigma_a = E_a \varepsilon = (70 \times 10^9)(1.3393 \times 10^{-3}) = 93.75\ MPa$

0.5]



$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu(\sigma_y + \sigma_z))$$

$$= \frac{1}{10 \times 10^3}(10 - 0.33(-15 - 26))$$

$$\varepsilon_x = 2.35 \times 10^{-3}$$

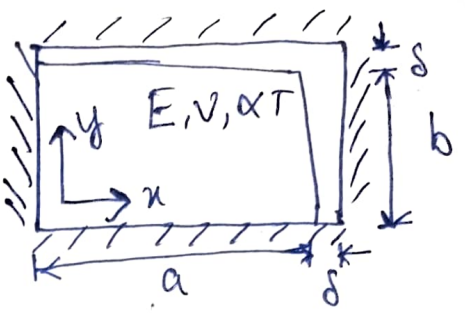
$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu(\sigma_x + \sigma_z)) = \frac{1}{10 \times 10^3}(-15 - \nu(10 - 26))$$

$$\varepsilon_y = -0.972 \times 10^{-3}$$

$$\varepsilon_z = \frac{1}{E}(\sigma_z - \nu(\sigma_x + \sigma_y)) = \frac{1}{(10)10^3}(-26 - 0.33(10 - 15))$$

$$\varepsilon_z = -2.44 \times 10^{-3}$$

Q6]



a) For $\Delta T < \Delta T_a$

The Plate is not confined in x & y direction

$\therefore \sigma_x = \sigma_y = 0$

$\therefore \varepsilon_x = \varepsilon_y = \alpha_T \Delta T$ from Hookes law

The right side space closes if $\varepsilon_x = \frac{\delta}{a}$

$$\therefore \boxed{\Delta T_a = \frac{\delta}{a \alpha_T}}$$

b) at a temp increase $\Delta T_a \leq \Delta T \leq \Delta T_b$ the plate can only expand freely in y-direction, while Strain in x-direction remains constant.

with $\sigma_y = 0$, & $\varepsilon_x = \frac{\delta}{a}$

$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu(0)}{E} + \alpha_T \Delta T \Rightarrow \frac{\sigma_x}{E} + \alpha_T \Delta T = \frac{\delta}{a}$$

$$\varepsilon_y = -\nu \frac{\sigma_x}{E} + \alpha_T \Delta T$$

The upper space closes if $\varepsilon_y = \frac{\delta}{b}$

$$\therefore \Delta T_b = \frac{\delta}{\alpha_T a} \quad \therefore \boxed{\Delta T_b = \frac{\delta}{\alpha_T a} \frac{a + \nu b}{(1+\nu) b}}$$

$$\boxed{\sigma_x = -\frac{E}{1+\nu} \frac{\delta (a-b)}{a b}}$$

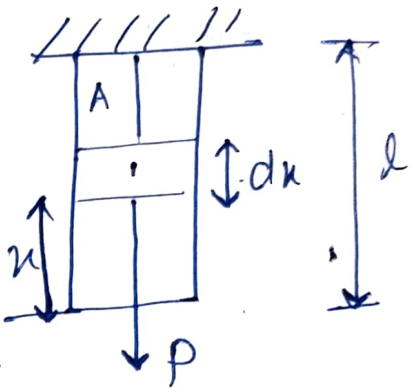
c) $\Delta T > \Delta Tb$, strains in both direction remain constant.

$\varepsilon x = \delta/a$, $\varepsilon y = \delta/b$

$E\varepsilon x = \sigma x - \nu\sigma y + E\alpha_T\Delta T$, $E\varepsilon y = \sigma y - \nu\sigma x + E\alpha_T\Delta T$

$$\sigma x = E\left[\frac{\delta(\nu a+b)}{(1-\nu^2)ab} - \frac{\alpha_T\Delta T}{1-\nu}\right], \sigma y = E\left[\frac{\delta(\nu b+a)}{(1-\nu^2)ab} - \frac{\alpha_T\Delta T}{1-\nu}\right]$$

Q.7] $\gamma \rightarrow$ weight density



we have,

$$\sigma = \frac{P}{A} + \gamma x$$

From Hookes Law, $\sigma = \varepsilon E$, $\varepsilon = \frac{\sigma}{E}$

$$\Delta = \int_0^L \varepsilon(x)\,dx$$

$$\Delta = \frac{1}{E}\int_0^L \left(\frac{P}{A} + \gamma x\right)dx = \frac{1}{E}\left(\frac{PL}{A} + \frac{\gamma L^2}{2}\right)$$

$$\Delta = \frac{PL}{AE} + \frac{\gamma L^2}{2E}$$